JOB PORTAL

Complete Project Documentation

Product scope • Features • Workflows • Technology • Security • Budget • Timeline • Deployment

PROJECT TYPE	Android job-seeker application + employer web portal + admin web portal
DELIVERY MODEL	Single-developer implementation
TARGET TIMELINE	Approximately 18 weeks (4–5 months)
PROPOSED HOSTING	Hostinger VPS for web dashboards and backend API
DOCUMENT VERSION	1.0 | 03 November 2026
STATUS	Proposed plan for client review and approval

PROJECT PURPOSE

Enable job seekers to discover and apply for jobs free of charge, allow approved employers to purchase subscriptions and publish vacancies, and give the platform owner centralized control through a secure admin dashboard.

Prepared as a consolidated implementation plan. Final scope, commercial terms, policies and third-party account ownership must be approved by the client before development begins.

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Source basis: the client-provided 10-page Job Portal Mobile Application scope document, consolidated with the final technical architecture and budget discussed for this project. Items not yet decided are marked as pending client confirmation.

1. Executive summary

The proposed solution is a job portal consisting of three user-facing interfaces connected to one shared backend: a Flutter Android app for candidates, a React.js web dashboard for employers, and a React.js web dashboard for administrators. Employers pay for subscription plans to publish jobs. Candidates can search and apply for jobs without paying.

The backend is a Node.js + Express.js API. Neon PostgreSQL stores structured platform data, Cloudflare R2 stores private files, Firebase Cloud Messaging delivers push notifications, and Razorpay processes employer subscription payments. Hostinger VPS is proposed for hosting the two web dashboards and backend API.

2. Project objectives and user roles

- **Candidate / Job seeker:** register, maintain a professional profile, upload a resume, discover jobs, apply and track application progress.
- **Employer / HR:** register a company, complete verification, buy a plan, publish vacancies within quota and manage applicants.
- **Admin / Owner:** manage candidates, employers, jobs, subscriptions, payments, content, settings, reports and staff permissions.
- Maintain strict separation between roles. Candidate accounts must not access employer or admin features.

Role	Access channel	Commercial model
Candidate	Android app	Free
Employer / HR	Responsive web dashboard	Paid subscription
Admin / Owner	Secure responsive web dashboard	Internal administration

3. Candidate Android application

Platform: Flutter + Dart. The candidate experience is free to use.

3.1 Account and authentication

- Mobile number registration and OTP verification.
- Email and password login option, if approved.
- Forgot password and account recovery.
- Accept Terms of Use and Privacy Policy.
- Secure logout and account deletion request.
- Rate limiting for OTP requests and login attempts.

3.2 Candidate profile and resume

- Personal details: name, mobile, email, location and profile photo.
- Career details: education, experience, skills, current salary and expected salary.
- Preferred role, job category and preferred locations.
- Upload, replace and delete resume files (PDF, DOC or DOCX).
- Profile completion indicator and profile editing.

3.3 Job discovery and search

- Home screen with latest, recommended, featured and recently viewed jobs.
- Search by job title, keyword, company, skill or location.
- Filters: location, category, experience, salary, job type, education and date posted.
- Job details: description, responsibilities, requirements, salary, benefits, vacancies and deadline.
- Company information, save/favourite jobs, share job links and report suspicious listings.

3.4 Applying and application tracking

- Apply using saved profile and resume.
- Employer screening questions when enabled.
- Prevent duplicate applications to the same job.
- My Applications page with submitted, viewed, shortlisted, interview, rejected, hired and withdrawn statuses.
- Allow candidate withdrawal where permitted by the platform policy.

3.5 Notifications and support

- In-app notifications with read/unread state and preferences.
- Push alerts for relevant job and application events.
- Help and support, FAQs, feedback and report-a-problem flow.

4. Employer web dashboard

Platform: React.js responsive web application.

4.1 Employer onboarding and company verification

- Employer registration using HR contact details, email and mobile verification.
- Company profile: name, logo, industry, website, address and description.
- Upload verification documents if required by the admin policy.
- Verification states: Pending, Approved, Rejected or Suspended.
- Only approved employers with an active subscription and available quota can publish jobs.
- Multiple HR members per company (Owner, HR Manager, Recruiter) are optional and require confirmation.

4.2 Employer overview dashboard

- Show active plan, start date, expiry date and subscription status.
- Show total, used and remaining job-post quota.
- Quick actions: Post a Job, Manage Jobs, Applicants, Company Profile, Subscription, Billing and Support.
- Show job views, applications and shortlisted candidates where tracking is available.
- Display low-quota and plan-expiry warnings.

4.3 Job posting and management

- Capture title, category, description, responsibilities, skills and qualifications.
- Capture experience, salary range, location, job type, vacancies and application deadline.
- Work mode: on-site, remote or hybrid.
- Save as Draft without consuming quota.
- Publish only after checking employer approval, active plan, valid plan dates and remaining quota.
- Edit, pause, close, renew or delete jobs.
- Job states: Draft, Pending Approval, Published, Paused, Closed, Expired and Rejected.

4.4 Applicant management

- View applicants for each job and filter by date, status, experience, qualification and location.
- Open candidate profiles and authorised resumes.
- Update application status: viewed, shortlisted, interview, rejected or hired.
- Add private HR notes and record interview date, time and instructions.

- Export applicant list to CSV/Excel if enabled by the client.

4.5 Subscription and billing

- View available plans, price, tax, duration and quota.
- Pay online through Razorpay.
- View invoices/receipts, payment history and subscription history.
- Renew, upgrade or change plan according to approved business rules.
- Block publishing when subscription is expired, suspended or out of quota.

5. Admin web dashboard

Platform: React.js responsive web application. Admin permissions must be enforced by the backend, not only by hiding UI elements.

5.1 Overview and analytics

- Dashboard totals for candidates, employers, jobs, applications, subscriptions and payments.
- Charts, date filters and recent activity.

5.2 Candidate management

- Search and view candidate accounts and profiles.
- Activate, suspend or block accounts.
- Handle complaints and account deletion requests.
- Never expose passwords, OTP values or payment secrets.

5.3 Employer management

- Review company details and verification documents.
- Approve, reject, suspend or reactivate employers.
- View employer plan, quota and payment history.
- Manually assign or extend plans only with a recorded reason.

5.4 Plans and quota management

- Create, edit, activate or deactivate subscription plans.
- Configure plan name, price, tax, duration and job-post quota.
- Manage coupons, trials and complimentary access if approved.

5.5 Job and content moderation

- Approve, reject, edit, pause or remove job listings.
- Manage job categories, skills and locations.
- Manage banners, featured jobs, FAQs, Terms and Privacy Policy.

5.6 Payments and finance

- View transactions and payment status.
- Review failed payments, refunds and disputes.
- Generate payment reports and exports.

5.7 Settings, staff and audit

- Staff roles: Super Admin, Operations, Content Moderator and Finance, if required.
- Manage notification templates and SMS/email/Firebase settings.

- Maintenance mode and system settings.
- Audit logs for sensitive admin actions.

6. Subscription plans and quota rules

Plan	Duration	Included job posts
Monthly	1 month	10
Quarterly	3 months	50
Yearly	12 months	150

- Quota is tracked per employer and per subscription period.
- One job consumes one quota unit at its first successful publication.
- Draft creation and editing an already-published job do not consume additional quota.
- Publishing is blocked when quota is exhausted or the subscription is not active.
- Unused quota does not carry forward under the proposed default policy.
- Closing or deleting a job does not restore consumed quota.
- Admin quota corrections must record the reason and the administrator responsible.
- On plan expiry, new publishing is blocked. Existing live jobs should be paused or closed by default, pending client approval.
- Renewal behaviour (immediate reset, extension or queued renewal) must be confirmed before implementation.

7. End-to-end workflows

7.1 Employer and job publication workflow

Step	Action	System behaviour
1	Employer registers	Verify contact details; employer remains Pending.
2	Company review	Admin approves or rejects company profile.
3	Employer buys plan	Payment is verified server-side; subscription becomes Active and quota is assigned.
4	Employer creates job	Job is saved as Draft; no quota is consumed.
5	Employer publishes	Backend checks approval, plan dates and quota; one quota unit is reserved/consumed atomically.
6	Admin moderation (if enabled)	Admin approves or rejects the listing; employer receives a reason.
7	Candidate applies	Application is stored; employer receives a notification.
8	HR reviews	Employer updates status; candidate receives a status notification.
9	Job closes/expires	New applications stop; application history remains available.

7.2 Payment workflow

- Backend creates a Razorpay order using the selected plan and server-calculated amount.
- Candidate/employer client is redirected to the payment experience; card details are handled by the gateway.
- Backend verifies payment signature and/or processes the trusted Razorpay webhook.
- Webhook processing is idempotent: duplicate callbacks cannot activate the plan or grant quota twice.
- Only verified successful payments activate or extend a subscription.
- Failed, pending or disputed payments do not grant quota. Payment and subscription states are stored separately.

7.3 Application status workflow

Submitted → Viewed → Shortlisted → Interview → Hired or Rejected. Candidate withdrawal is supported subject to policy. The platform records timestamps and status history for audit and user visibility.

8. Notifications and communication

Event	Recipient	Example purpose
New account / OTP	Candidate or employer	Verification code or account confirmation.
New application	Employer	A candidate applied to a vacancy.
Application status change	Candidate	Viewed, shortlisted, interview, rejected or hired.
Payment result	Employer	Payment success/failure and receipt reference.
Quota warning	Employer	Quota is low or exhausted.
Plan expiry	Employer	Subscription is nearing expiry or has expired.
Job moderation	Employer	Job approved, rejected, paused or removed, with reason.
Admin alerts	Admin	New approvals, reported jobs and payment issues.

- Firebase Cloud Messaging (FCM) provides push notifications to the Android app.
- Notifications should also be saved in the database/in-app inbox so missed push messages are not lost.
- SMS is used for OTP and any agreed transactional updates; email is used for receipts and important messages.
- WhatsApp notifications are optional and require a separately approved provider and budget.

9. Payments and billing

- Use one client-approved payment gateway; Razorpay is the proposed option.
- Verify payment outcomes on the backend, not from the mobile app success screen alone.
- Store gateway order ID, payment ID, amount, currency, status and relevant timestamps.
- Support invoices/receipts and payment/subscription history.
- Never store card numbers or CVV.
- GST treatment, invoice format, refund policy, auto-renewal and merchant account ownership are client decisions.

10. Advertising and monetization

Google AdMob is proposed for advertisements inside the candidate Android app. Google AdSense is a separate optional possibility for web content, but it is not required for the first release.

Ad format	Proposed placement	Rules
Banner	Bottom of home, search results and job-list screens.	Must not cover navigation or action buttons.
Native	Between job cards after a configurable number of listings.	Clearly labelled “Ad”; must not look like a job listing.
Interstitial	After an application is submitted or after several job views, subject to policy.	Frequency-capped; never interrupt login, OTP, payment or application entry.
Rewarded	Future enhancement only.	Not included in first release.

- Ads are shown to job seekers only; employer and admin dashboards remain ad-free under the proposed policy.
- No ads on login, OTP, payment, resume upload or application form screens.
- Use Google test ads during development; activate live ads only after approval and release.

- Client owns the AdMob account and payment profile and provides ad unit IDs.
- Privacy policy and Google Play Data Safety declarations must disclose applicable advertising data use.

11. Technology stack and architecture

Layer	Selected technology	Responsibility
Candidate mobile app	Flutter + Dart	Android user interface and device integration.
Employer web dashboard	React.js	Employer onboarding, subscription and job/applicant management.
Admin web dashboard	React.js	Administration, moderation, plans, payments and reports.
Backend API	Node.js + Express.js	Authentication, business rules, permissions, workflows and integrations.
ORM	Prisma	Database schema, migrations and typed data access.
Relational database	Neon PostgreSQL	Users, profiles, companies, jobs, subscriptions, payments and applications.
File storage	Cloudflare R2	Private resumes, profile photos, company logos and verification documents.
Hosting	Hostinger VPS	Serve web applications and run the backend API.
Push notifications	Firebase FCM	Push delivery to Android devices.
Payment gateway	Razorpay	Employer subscription payments.
Code repository	GitHub	Source control, code review and release history.

- Use one shared backend API for all three interfaces.
- Employer and admin dashboards may be deployed as one React application with separate routes and backend-enforced role permissions.
- Use Nginx as a reverse proxy and a process manager (for example, PM2 or systemd) to keep the Node.js service running on the VPS.
- Use HTTPS, environment variables and restricted server access. Never commit API keys or secrets to GitHub.
- Neon and R2 are external managed services; their usage limits and billing must be monitored.
- VPS is a persistent server, unlike serverless function hosting. The development team is responsible for operating-system updates, firewall, deployment, monitoring and backups.

12. Database and core records

Record / entity	Purpose
Users	Login identity, role, status and contact verification.
Candidate Profiles	Personal, education, experience, skills, preferences and resume references.
Companies	Employer/company profile and verification status.
Employer Members	HR users linked to a company, where multi-user access is enabled.
Plans	Plan name, duration, price, tax and quota.
Subscriptions	Employer, plan, dates, assigned quota, used quota and status.
Payments / Invoices	Gateway references, amount, tax, payment status and receipt/invoice data.
Jobs	Job details, status, publication dates, deadline and quota reference.
Applications	Candidate, job, submitted resume snapshot and application status.
Notifications	Recipient, message, delivery metadata and read status.
Audit Logs	Who performed a sensitive action, when and why.

The exact schema, indexes, retention rules and migration strategy will be finalized during database design. Sensitive files should be stored in private object storage, not as public database URLs.

13. Security, privacy and reliability

13.1 Authentication and authorization

- Hash passwords using a modern password-hashing algorithm.
- Use short-lived access tokens and secure refresh-token handling where applicable.
- Enforce role and company-level permissions on every protected backend endpoint.
- Throttle OTP requests and login attempts; prevent account enumeration where practical.

13.2 Data and file protection

- Use HTTPS for all traffic.
- Validate and sanitize user input; use parameterized database access through Prisma.
- Store resumes and verification documents in private Cloudflare R2 buckets.
- Issue short-lived signed URLs only after authorization checks.
- Restrict file types and sizes; consider malware scanning for uploaded documents.

13.3 Payments and audit

- Verify Razorpay signatures and webhooks server-side.
- Use idempotency and database transactions for subscription/quota changes.
- Never log passwords, OTPs, payment secrets or sensitive document contents.
- Maintain audit records for admin decisions, quota changes, refunds and account suspensions.

13.4 Privacy and user rights

- Obtain acceptance of Terms of Use and Privacy Policy.
- Provide consent and advertising disclosures where required.
- Implement account deletion request handling and a documented data-retention policy.
- Limit candidate contact details and resume access to authorized employers and permitted workflow stages.

13.5 Reliability and operations

- Schedule database backups and file-storage backup/retention policies.
- Monitor application errors, server resources, database capacity and storage usage.
- Document restore procedures and test recovery periodically.
- Use pagination, indexes, optimized images and caching where appropriate.

14. Reports and analytics

Report	Suggested metrics
Candidates	Registrations, active/inactive users, location and profile completion.
Employers	Registrations, approval status, active plans and renewals.
Subscriptions	Plan sales, active/expired subscriptions and revenue.
Quota	Purchased, used, remaining and employers with zero quota.
Jobs	Created, published, closed, expired, category and location split.
Applications	Total applications, applications per job, status distribution and hiring outcomes.

Reports should include date filters. CSV/Excel export is available to authorized admin roles. Revenue reports should reconcile against verified payment records and refunds.

15. UI/UX and quality requirements

- Responsive employer and admin dashboards for agreed desktop, tablet and mobile browser sizes.
- Agree minimum Android version and test on representative Android devices.
- Consistent loading, empty, success, error and validation states.
- Search results and large tables must use pagination and suitable database indexes.
- Prevent duplicate applications and duplicate subscription activation.
- Use accessible labels, readable typography, clear navigation and confirmation for destructive actions.
- Define uptime, performance and support targets in the service agreement rather than assuming a guarantee.

16. Development phases and timeline

Phase	Work items	Estimate
1. Discovery and design	Confirm requirements, user flows, UI/UX, database design and API contract.	2 weeks
2. Backend foundation	Authentication, roles, core schema, candidate/employer APIs and file upload.	3 weeks
3. Employer portal	Company verification, dashboard, job posting, applicant management and subscription screens.	3 weeks
4. Admin portal	User/employer/job moderation, plans, quota, payments, content and reports.	2 weeks
5. Android app	Candidate account, profile, resume, search, job details, apply, history and notifications.	4 weeks
6. Integrations	Razorpay, FCM, OTP/email provider, AdMob test integration.	2 weeks
7. QA and launch	End-to-end tests, security checks, production setup, Play Store submission support and handover.	2 weeks
TOTAL		18 weeks

Timeline assumes one developer with relevant experience working consistently, timely client feedback, and access to third-party accounts. App-store review delays, scope changes and provider approval times may extend delivery.

17. Deployment and domain plan

Domain / subdomain	Purpose
yourdomain.com	Optional public landing page / main website.
employer.yourdomain.com	Employer dashboard.
admin.yourdomain.com	Admin dashboard.
api.yourdomain.com	Backend API.

- Register the domain with Hostinger or another registrar selected by the client.
- Point DNS records to the Hostinger VPS public IP.
- Configure Nginx virtual hosts/reverse proxy for the dashboard and API subdomains.
- Install and renew TLS certificates (for example, Let's Encrypt).
- Deploy frontend build files and run the backend process under a process manager.
- Configure production environment variables, database connection, R2 credentials, Razorpay webhook URL and FCM credentials.
- Set up firewall, SSH key access, automated backups, log rotation and monitoring before go-live.

18. Infrastructure budget

The following is an initial annual planning estimate, not a provider quote. Actual cost depends on billing term, renewal price, region, storage, traffic, database usage, taxes and third-party consumption.

Service	Estimated annual allowance	Notes
Hostinger VPS	₹10,000–₹18,000	Choose a commercial VPS plan suitable for app + dashboards; confirm renewal price and resources.
Domain	₹900–₹1,500	Depends on domain extension and renewal pricing.
Neon PostgreSQL	₹0–₹5,000	Free tier may be sufficient for early use; paid usage may be needed as load grows.
Cloudflare R2	₹0–₹1,500	Free allowance may cover early usage; storage and operations can become chargeable.
Firebase FCM	Free for push delivery	Other Firebase products may have separate pricing.
GitHub	Free	Private repository and basic collaboration.
Razorpay	Variable	Transaction fees and applicable taxes; confirm current merchant pricing.
SMS/OTP	₹1,000–₹5,000+	Usage-based; varies by provider, route and volume.
Email service	₹0–₹2,000	Depends on provider and sending volume.
Google Play Console	US$25 one-time	Developer registration fee; account should be owned by client.

RECOMMENDED FIRST-YEAR INFRASTRUCTURE ALLOWANCE

₹20,000–₹30,000 as a planning allowance, including contingency. This is not a fixed or guaranteed price.

- Not included: developer salary/project development fee, marketing, ongoing maintenance, GST/taxes where applicable, payment gateway fees, optional WhatsApp provider, paid OTP/email overages and major future upgrades.
- Client should own the domain, hosting, database, storage, Firebase, Razorpay, AdMob and Play Console accounts wherever possible.

• Review actual usage and renewal pricing before the end of the first billing term.

19. Testing and acceptance criteria

• Candidate can register, complete profile, upload a resume, search jobs, apply and view application history.

• Duplicate applications to the same job are blocked.

• Employer cannot publish until company approval and an active subscription with remaining quota.

• Monthly, quarterly and yearly plans assign 10, 50 and 150 job posts respectively, after client approval of plan configuration.

• Drafts do not consume quota; first publication consumes one unit; edits do not consume extra quota.

• Publishing is blocked when quota is exhausted or the plan has expired.

• Payment success is verified server-side; failed or duplicate callbacks do not activate a plan twice.

• Employer can view applicants and update application status; candidate receives agreed status notifications.

• Admin can manage users, employers, jobs, plans, quota, payments, content and settings according to role permissions.

• Reports reconcile with stored records and authorized users can export agreed report data.

• Ads appear only in approved placements and can be disabled through the agreed admin control.

• Backups, restore process, production monitoring and deployment documentation are handed over.

20. Client decisions and dependencies

Decision needed	Proposed default / required input
Employer portal	Responsive web dashboard.
Admin portal	Responsive web dashboard; confirm whether staff roles are needed.
Plan pricing and GST	Client to provide monthly, quarterly and yearly prices, GST treatment and invoice details.
Quota on rejected/re-published jobs	Count once on first publication; confirm edge cases.
Unused quota	No carry-forward.
Plan expiry	Pause or close live jobs; confirm whether any grace period is required.
Renewal	Client to choose immediate reset, extension or queued renewal.
Job moderation	Client to decide whether every job requires admin approval.
Candidate contact visibility	Proposed: reveal only after application or shortlist; confirm.
OTP/email provider	Client to approve provider and usage budget.
Refund and auto-renewal	Client to provide written policy.
Minimum Android version	Client/development team to agree.
Ad placements	Proposed banner/native ads for candidates only; approve frequency and screens.
Account ownership	Client to create/own third-party service accounts and provide access securely.

21. Assumptions, exclusions and change control

• This document describes the proposed first release based on the supplied project scope and subsequent architecture decisions.

• Employer and admin interfaces are web dashboards; a separate employer/admin mobile app is not included.

• One shared backend API serves the Android app and both dashboards.

• Multi-HR company accounts are optional and must be confirmed.

- WhatsApp messaging, in-app chat, interview calendar integrations, AI job matching, resume parsing, advanced search infrastructure and courses/video modules are not included unless separately approved.
- Any new feature, workflow, screen, integration or third-party service outside the approved scope must be assessed for cost and timeline impact and approved in writing.
- Third-party service outages, policy changes, app-store approval decisions and usage charges are outside the developer's direct control.

Commercial terms such as development fee, milestone payments, GST, maintenance period, bug-fix warranty, support response times and service-level commitments should be recorded in a separate signed quotation or agreement.

22. Launch checklist

- Client approves scope, policies, plan prices and open decisions.
- Brand assets, app name, logo, content, job categories and locations are provided.
- Client-owned domain and hosting accounts are ready.
- Production database, storage, backups and monitoring are configured.
- Razorpay live account and webhook are verified.
- OTP and email providers are approved and tested.
- Firebase project and Android app configuration are ready.
- AdMob account, app registration and ad unit IDs are approved.
- Privacy Policy, Terms of Use and account deletion flow are published.
- Admin users and permissions are configured securely.
- Candidate, employer and admin end-to-end acceptance tests pass.
- Play Store listing, screenshots, Data Safety details and release build are prepared.
- Deployment, rollback, backup and restore instructions are handed over.
- Client signs off the production release.